Supplementary Methods

AHP-weighted composite decision scoring of TNBC exosomal proteomics candidates

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This document accompanies the main manuscript and expands Methods with full pipeline detail, parameter justification, and package versions for each stage (Scripts 00–14). It is provided for reproducibility.

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S1 Data acquisition and quality control

S1.1 Dataset and experimental design

Raw quantitative proteomics data were obtained from ProteomeXchange/PRIDE dataset `PXD056161`, originally generated as part of a study profiling exosomal cargo secreted by the non-tumorigenic breast epithelial line MCF-10A and the triple-negative breast cancer line MDA-MB-231 (Zaripov et al., 2025). The present work re-uses only the nanoLC-MS-based exosomal *proteomics* arm of that deposition (its accompanying CE-MS metabolomics data on purine/carnitine metabolism are not used here). Samples are organized as two conditions – Normal (MCF-10A-derived) and Tumor (MDA-MB-231-derived) – with $n = 3$ biological replicates per condition (6 samples total), consistent with the per-group coefficient-of-variation estimation used later for imputation (§2).

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S1.2 Gene/protein identifier mapping

Gene and protein identifiers were retrieved via the `biomaRt`/Ensembl BioMart R package (Durinck et al., 2009), connecting to the Ensembl US East mirror (`useast.ensembl.org`) and querying the `hsapiens_gene_ensembl` dataset (Ensembl release 116, GRCh38; accessed 16 August 2026) (scripts 00). Representative UniProt Swiss-Prot accessions were extracted from each MaxQuant protein group (the first semicolon-delimited accession per group) and de-duplicated to form the query universe. `getBM()` was queried with filter `uniprotswissprot` against attributes `uniprotswissprot`, `external_gene_name`, `description`, `go_id`, `name_1006` and `namespace_1003`, yielding a per-protein table of HGNC gene symbol, textual description, and GO term/name/domain (BP/MF/CC) annotations used to harmonize MaxQuant protein groups with downstream pathway and network resources.

S1.3 MaxQuant intake and quality control

MaxQuant search output was taken directly from the PRIDE deposition (i.e. not re-searched in-house), using MaxQuant version 2.3.0.0; `proteinGroups.txt` served as the intake file. Two contaminant-removal steps are applied independently, at different pipeline stages, and are not duplicates of one another: (i) here, standard MaxQuant-flagged contaminant and reverse-hit entries are removed as part of intake QC; (ii) a separate, EV-biology-aware filter (MISEV-guideline histone/keratin/actin/tubulin regex plus a manual blacklist, §5) is applied much later, after GSEA, to the post-hoc core-enrichment candidate pool rather than to the whole protein-group table.

S1.4 LFQ intensity processing

Label-free quantification (LFQ) intensities were extracted per protein group and $\log_2$-transformed prior to all downstream statistical steps. No zero-substitution or imputation is applied at this stage: values absent from the search output remain `NA` in the resulting `log_lfq_matrix`, which is the direct input to the missingness diagnostics of §2, where it is reshaped into a long-format protein$\times$sample table for statistical modeling of the missingness mechanism.

S1.5 Sample-level QC and PCA

Sample-level quality was assessed via principal component analysis (run as part of script 05, alongside the imputation-stability check of §2.5, rather than as a separate early-pipeline step), used to check (i) whether any individual sample fails to cluster with its condition group, (ii) the overall variance structure across the dataset, and (iii) a rough read on whether the biological signal of interest (tumor vs. normal) accounts for a relatively large share of that variance. This QC pass did not flag any sample for exclusion – all 6 samples were retained downstream. The same before/after-imputation PCA panel (Fig. S1E) serves both this general sample-level QC role and the imputation-fidelity role described in §2.5.

S2 Missingness diagnostics and imputation

S2.1 Technical-replicate collapsing

Where a biological replicate was measured across multiple technical- replicate columns, intensities were collapsed by taking the per-protein row mean across those columns (`na.rm=TRUE`), yielding one column per biological replicate (script 04). This collapsed matrix is the input to all subsequent missingness and filtering steps.

S2.2 Detection filter

Proteins were retained only if detected (non-`NA`) in at least 2 of the 3 biological replicates within *either* condition (Normal OR Tumor); proteins meeting this bar in neither condition were dropped. This is the

"2/3-missing filter" referenced in the main-text pipeline overview (Methods §4).

S2.3 Missingness-mechanism diagnostic

Prior to filtering, the full (uncollapsed) log-intensity matrix was reshaped to long format (protein × sample) and each observation coded as detected/not-detected. A logistic regression of detection status on per-protein mean log-intensity,

$$\operatorname{logit} P(\text{Detected}_{ij} = 1) = \beta_0 + \beta_1 \overline{\text{Intensity}}_i,$$

was fit to test for abundance-dependent detection. The fit showed a strong positive association between abundance and detection probability ($\hat{\beta}_1 = 0.43$, $p < 2 \times 10^{-16}$), consistent with a limit-of-detection-driven MNAR (missing-not-at-random) mechanism rather than missingness independent of abundance. This relationship is shown as a jittered detection-vs-intensity scatter with fitted binomial curve in Fig. S1B.

S2.4 Extreme (3-vs-0) missingness pattern

Within the post-filter protein set, proteins detected in all 3 replicates of one condition and none of the 3 replicates of the other (a complete "3v0" presence/absence swap between Normal and Tumor) were flagged and counted separately in each direction (Normal-only-3v0, Tumor-only-3v0). This pattern was observed in approximately 17.7% of the filtered protein set. Given the abundance-dependent detection mechanism established in §2.3, these 3v0 cases are interpreted as likely reflecting detection-limit censoring of a genuinely lower-abundance protein rather than true qualitative (on/off) biological absence – motivating the decision (§2.5) to impute these values under the same group-wide MNAR model as any other missing value, rather than hard-coding them as structural zeros. (This interpretation is revisited as an explicit Limitation regarding condition-specific near-binary missingness; no wet-lab validation, e.g. targeted MS or Western blot, was performed to confirm true qualitative absence for any individual 3v0 candidate.)

S2.5 Imputation

Remaining missing values were imputed *per condition group* (Normal and Tumor imputed independently, script 05, `set.seed(30032026)`) using a left-shifted, width-narrowed Gaussian model in the style of Perseus (Tyanova et al., 2016): within each group's collapsed intensity matrix, a single mean μ and SD σ were estimated from all observed (non-missing) values pooled across every protein and replicate in that group, and each missing cell in the group was independently drawn from $\mathcal{N}(\mu - 1.8\sigma,\ (0.3\sigma)^2)$ ($d = 1.8$ downshift, $w = 0.3$ width). Imputation is therefore group-global rather than per-protein or per-sample-column: all missing values within a condition share the same source distribution, positioned 1.8σ below that condition's own overall mean. This differs from Perseus's typical per-column default. The per-group vs. pooled-across-both-groups choice itself is supported by the earlier per-group coefficient-of-variation estimates (mean/SD = 0.51%, median/MAD = 0.71% across the 6 samples, see Fig. S1D): the two condition groups have small but non-identical intensity-distribution variance, so a single pooled-across-groups Gaussian was judged to understate that condition-specific structure – the original-vs-imputed distribution boxplot (Fig. S1C) visually supports this group-wise treatment, showing each condition's imputed values tracking that condition's own observed distribution rather than a shared one.

Two internal checks were run on the resulting imputed matrix: (i) a distribution comparison of original vs. imputed log-intensity values, available as both a boxplot (Fig. S1C); (ii) a PCA-structure stability check, comparing a complete-case-only PCA (proteins with zero missingness, scaled) against a PCA on the fully imputed matrix, via the Spearman correlation between PC1 gene-loading vectors before and after imputation ($\rho = 0.9994$, i.e. near-perfect preservation of the PC1 loading structure post-imputation) – shown as the two-panel before/after PCA of Fig. S1E, which doubles as the general sample-level QC

PCA of §1.5. Post-imputation robustness against the complete-case (non-imputed) subset is assessed separately in §3.

S3 Differential expression and robustness check

S3.1 Differential expression model

Differential expression was assessed with `limma` (script 06). A no-intercept design matrix was used, `model.matrix(~0+group)`, with `group` coded Normal/Tumor ($n = 3$ each), and a single contrast Tumor – Normal was fit via `lmFit()` → `contrasts.fit()` → `eBayes()` (standard empirical-Bayes moderated t-statistics) on the full imputed intensity matrix. The complete, unfiltered ranked table (`topTable(..., number=Inf, sort.by="P")`) was retained as the working differential-expression result – no p- or FDR-threshold was applied at this stage, consistent with the FDR-treated-continuously policy used throughout the candidate-scoping pipeline (§4–§5). Row identifiers were reduced to a representative UniProt accession per protein group (first semicolon-delimited entry). Raw p-value and adjusted-p (FDR) distributions were compared visually (overlaid histograms) as a diagnostic check for anti-conservative behavior.

S3.2 Complete-case robustness comparison

As an internal check on the influence of imputation (§2.5), the identical limma model (same design, same contrast) was re-fit on an independent, non-imputed subset: proteins with *zero* missing values across all 6 samples (`complete_inx`) taken directly from the collapsed (pre-imputation) matrix.

Robustness was assessed purely via continuous concordance, with no significance threshold applied at any point – consistent with this paper's general treatment of FDR as a continuous quantity throughout, not a hard gate. Both result tables were first restricted to their common protein set (i.e. the complete-case set), and two Pearson correlations were computed on that shared set between the imputed-pipeline and complete-case fits (script 07): the correlation of moderated t-statistics (`cor_t`) and the correlation of $-\log_{10}(\text{adj.}P)$ values (`cor_FDR`). A separate, likewise threshold-free coverage statistic was also computed – `coverage_ratio` = |common set|/|full imputed-analysis protein set| – quantifying what fraction of the analyzed protein universe required no imputed values at all, kept distinct from (rather than conflated with) the two concordance correlations.

Concordance was high: `cor_t` = 0.998 and `cor_FDR` = 0.996 on the $n = 353$ complete-case proteins common to both fits, indicating that imputation does not materially distort either the direction/magnitude of the moderated t-statistic or the FDR ordering for proteins that were already fully observed. The complete-case set (353 of 866 proteins in the full imputed-analysis universe, `coverage_ratio` = 0.408) represents proteins requiring no imputed values at all; the remaining ∼59% of the analyzed pool required at least one imputed value, for which this complete-case comparison cannot directly speak but for which the group-level imputation-fidelity check of §2.5 (PC1 correlation $\rho = 0.999$) provides indirect support.

S4 Pathway enrichment (GSEA)

S4.1 Ranked gene list construction

Starting from the full (unfiltered, imputed-pipeline) differential expression table of §3.1, representative UniProt accessions were extracted per protein group (script 08, same first-accession convention as §1.2/§3.1) and mapped to Entrez gene IDs via `clusterProfiler::bitr()` against `org.Hs.eg.db`. Where this mapping produced duplicate Entrez IDs (multiple protein groups collapsing to the same gene), only the entry with the largest $|t|$ (moderated t-statistic) was retained per gene. The resulting pre-ranked gene list used the moderated t-statistic itself as the ranking metric (not $\log_2$FC or $-\log_{10}P$), sorted decreasing, with any remaining `NA` entries dropped.

S4.2 Reactome GSEA

Gene set enrichment analysis against Reactome pathways was run with `ReactomePA::gsePathway()` (`organism="human"`, `minGSSize=10`, `maxGSSize=500`, `pvalueCutoff=0.05`), with both a global `set.seed(30032026)` *and* `seed=TRUE` passed to `gsePathway()` itself for permutation reproducibility (Subramanian et al., 2005; Yu and He, 2016). `pvalueCutoff=0.05` is applied to the *adjusted* p-value internally by `gsePathway()`, so the returned result object already contains only pathways with $q < 0.05$ – this is a tighter, explicitly chosen cutoff, not the classic Broad GSEA default of $q < 0.25$.

S4.3 Supplementary magnitude filter

A further filter, $|\text{NES}| > 1.5$ AND `p.adjust` < 0.05, was applied to the returned pathway table. The `p.adjust` < 0.05 clause is redundant with §4.2's `pvalueCutoff=0.05` (every row already satisfies it by construction) and is retained in the filtering code only for clarity, not because it removes any additional rows. The effective, non-redundant filter is therefore the normalized-enrichment-score magnitude criterion $|\text{NES}| > 1.5$ alone, applied as an author-chosen supplementary stringency criterion on top of GSEA's own significance threshold. Applied to the real data: 159 pathways satisfied $q < 0.05$ (`reactome_gsea_full.csv`), of which 153 additionally satisfied $|\text{NES}| > 1.5$ (`reactome_gsea_filtered.csv`) – i.e. the magnitude filter removed 6 statistically-significant but lower-effect-size pathways from the working pathway set carried forward to §5.

S5 Core-enrichment gene pool and contaminant filtering

S5.1 Core-enrichment gene extraction

The `core_enrichment` column of the 153 NES-filtered Reactome pathways (§4.3) – the leading-edge Entrez gene IDs identified by `gsePathway()` as driving each pathway's enrichment score – was split on `"/"` and pooled into a single de-duplicated candidate gene list (script 09). This produced **193 unique Entrez gene IDs**, the working candidate universe carried into the contaminant filtering below and, subsequently, into STRING network construction (§6).

S5.2 Re-annotation to UniProt/HGNC

This Entrez-keyed candidate list was mapped back to UniProt/HGNC identity via a second, independent `biomaRt` query (distinct from the intake-stage mapping of §1.2): `getBM()` with filter `entrezgene_id` against attributes `entrezgene_id`, `hgnc_symbol`, `uniprotswissprot`, queried against a mirror-failover Ensembl connection (`asia`, `useast`, `uswest`, `www` mirrors tried in order) for query robustness. Rows with a missing or blank `uniprotswissprot` were dropped at this step. The result was then joined (by `uniprotswissprot`) against a pre-computed GO/description annotation table (`annotation_raw.csv`) to attach gene symbol, textual description, and GO term name/ID triplets (biological process, cellular component, molecular function) per protein.

S5.3 Contaminant filtering

Two independent contaminant-removal criteria were applied to this annotated candidate table:

(i) a case-insensitive regular-expression filter, `"histone|keratin|actin|tubulin"`, applied *separately* to both the protein textual description field and the gene symbol field (i.e. a candidate is dropped if either field matches), targeting proteins with strong evidence of being non-vesicular, co-isolated background rather than genuine EV cargo. Histone- and keratin-family proteins follow MISEV2023's explicit classification of nuclear (histone) and cytoskeleton/other (cytokeratin) proteins as markers of non-EV co-isolated material (Welsh et al., 2024). Actin- and tubulin-family

proteins are flagged on more recent, EV-specific evidence: density-gradient protein-correlation-profiling data show that abundant cytosolic proteins detected in small-EV preparations, including specific actin (ACTG1) and tubulin (TUBA4A) isoforms and the actin-nucleating ARP2/3 complex, are not encapsulated within vesicles but are externally associated, most likely originating from co-purifying cellular debris and protein aggregates rather than selectively sorted cargo (Morales-Sanfrutos et al., 2026) — a finding that also revises MISEV2023's own classification of actin/tubulin as EV-associated cytosolic markers (this is deliberately not a CRAPome-style filter, which targets sample-handling contaminants rather than EV-co-isolation background);

(ii) a manual 6-gene blacklist (`FMNL1`, `H2AX`, `H4C6`, `H3C1`, `H3-3B`, `H2AZ2`), matched exactly against the gene symbol field. The five histone-variant genes extend the regex-based histone filter to gene symbols following current HGNC nomenclature, which the `"histone"` pattern does not match. FMNL1 was added on a mechanistic rationale rather than gene-symbol matching: as an FH2-domain actin-nucleating/elongating protein, it is functionally analogous to the actin-associated complexes (e.g. ARP2/3) recently shown to co-purify with small-EV preparations as non-vesicular background (Morales-Sanfrutos et al., 2026), and was flagged by manual inspection accordingly rather than by any contaminant database cross-check.

Rows with any remaining missing value in the annotated table were then dropped (`na.omit()`) – a further, non-regex source of attrition that removes candidates with incomplete GO annotation, not just contaminant matches. The real post-filter output (`annotated_gene_pool.csv`) contains **152 candidates** – i.e. this stage's combined attrition (UniProt mapping loss, both contaminant criteria, and `na.omit()` together) removes 41 of the 193 starting core-enrichment genes; ; of these, 17 were removed by the two contaminant criteria (Fig. S1F breakdown: 7 histone, 5 actin, 5 tubulin, 0 keratin and no overlap between categories.)

S6 STRING network construction

S6.1 Network query and summary

HGNC gene symbols from the filtered candidate pool (`annotated_gene_pool.csv`, §5.3, 152 candidates) were passed to a project-local helper function, `build_string_network()` (script `string_api_helper.R`), which queries the STRING REST API and returns per-gene network summary statistics (script 10). The returned table was joined back to the candidate annotation (by `Symbol` ↔ `hgnc_symbol`) and reduced to `Symbol`, `uniprotswissprot`, `degree`, and `betweenness` per candidate.

Gene symbols are first resolved to STRING identifiers (species `9606`, human; best/first match per symbol) via STRING's `get_string_ids` endpoint, then the interaction network is queried via the `network` endpoint restricted to only those resolved candidate IDs – i.e. the network is built strictly among the candidate pool itself, not as a lookup against the full STRING interactome (candidate-subnetwork-only). `required_score=700` (STRING's 0–1000 combined-score scale, corresponding to the standard "high confidence" 0.7 threshold) is passed to the API call and then re-applied client-side on the returned edge scores as a redundant double-check. The resulting edge list is used to build an undirected graph (`igraph`, simplified to remove duplicate edges/self-loops), from which `degree()` and `betweenness()` are computed. Because the summary table is built from `igraph::degree()`/ `betweenness()` on that graph object, any candidate with *zero* edges at ≥ 700 confidence within the candidate subnetwork simply never appears as a node in the graph and is therefore absent from the returned summary table entirely – it is dropped, not zero-filled. The helper also defines an explicit fallback (returning degree = betweenness = 0 for every input gene) for the edge case where the STRING query returns zero edges for the entire input set – a code path present for robustness but not expected to trigger given the real candidate pool's connectivity.

S7 AHP-weighted composite decision scoring (AHP-CDS)

S7.1 Saaty pairwise comparison and weight derivation

Criterion weights were derived from a 4×4 Saaty pairwise comparison matrix (script 12, criteria ordered FC, FDR, Betweenness, Degree in the matrix itself):

$$M = \begin{pmatrix} 1 & 4 & 5 & 2.5 \\ 1/4 & 1 & 2 & 1 \\ 1/5 & 1/2 & 1 & 1/5 \\ 1/2.5 & 1 & 5 & 1 \end{pmatrix}$$

via column-normalization followed by row-averaging (the standard Saaty eigenvector approximation): each column of M is normalized to sum to 1, and each criterion's weight is the mean of its row in the normalized matrix. This yields $w_{\text{FC}} = 0.5181$, $w_{\text{FDR}} = 0.1639$, $w_{\text{Bet}} = 0.0761$, $w_{\text{Deg}} = 0.2419$. Consistency was assessed via $\lambda_{\max} = \text{mean}(Mw/w) = 4.1358$, $\text{CI} = (\lambda_{\max} - n)/(n-1) = 0.0453$, and $\text{CR} = \text{CI}/\text{RI} = 0.0503$ (Random Index $\text{RI} = 0.90$ for $n = 4$), well under the conventional $\text{CR} < 0.1$ acceptability threshold (Saaty et al., 2022).

S7.2 Composite Decision Score

Each of the four criteria was independently normalized to $[0,1]$ via min–max scaling, with two criteria pre-transformed before scaling: FDR via $-\log_{10}(\max(\text{adj.}P, 10^{-10}))$ (numerical floor to avoid $-\log_{10}(0)$), and degree/betweenness each via $\log_1 p(\cdot)$ (to compress their right-skewed, scale-free-network distributions before min–max scaling); fold-change was min–max-scaled on $|\log \text{FC}|$ (magnitude, direction-agnostic). The Composite Decision Score is the resulting weighted linear sum,

$$\text{CDS}_i = w_{\text{FC}}\, n_{\text{FC},i} + w_{\text{FDR}}\, n_{\text{FDR},i} + w_{\text{Bet}}\, n_{\text{Bet},i} + w_{\text{Deg}}\, n_{\text{Deg},i},$$

computed over the final 143-candidate DE+network-integrated table (§6.1) and sorted descending. A final `distinct(UNIPROT, .keep_all=TRUE)` step retains, for any UniProt accession appearing more than once in that table, only its highest-CDS occurrence (since deduplication is applied after sorting).

S8 Random Walk with Restart (RWR) comparator

S8.1 Algorithm

The RWR/Global Ranking (GR) comparator (script `07_ren2019_gr.R`) is a direct re-implementation of the heat-kernel diffusion algorithm from Ren *et al.* (Ren et al., 2019), reproduced from their published MATLAB source (`GR_DEPs.m`):

$$L = I - D^{-1}A, \qquad M = I - \tfrac{\alpha}{N}L, \qquad K = M^N \ (\approx \exp(-\alpha L)), \qquad p = p_0 K,$$

with L the random-walk-normalized graph Laplacian, p_0 a binary (un-normalized) seed indicator vector, and $\alpha = 0.5$, $N = 3$ (kept verbatim from the cited protocol, not tuned). Three deviations from the original protocol are explicitly documented in the script and disclosed here rather than silently varied: (i) the network is this project's STRING subnetwork (combined_score > 0.7), binarized to match the unweighted property of Ren's HINT interactome – only that unweighted property is matched, not HINT's actual source/topology; (ii) the DEP seed set is defined via Bonferroni correction applied to this project's own limma p-values, rather than re-deriving DEPs via Welch's t-test as in the original paper – chosen to keep one DE pipeline across the whole manuscript while matching Ren's correction stringency, and, as in the original, using the full Bonferroni-significant list with no top-N cutoff; (iii) $N = 3$ is a coarse truncation of $\exp(-\alpha L)$, kept exactly as reported rather than tuned, with no sensitivity check at larger N performed.

S8.2 Seed set and network construction

Two distinct DE-significance thresholds are used at two different points in this script, which should not be conflated: a BH-adjusted $p < 0.05$ subset of the full 866-protein imputed DE table (§3.1) is first used only to build the STRING query universe (resolved to STRING IDs in batches of 20 identifiers, with a 1-second pause between batches for API courtesy); the actual RWR seed set is the *stricter* Bonferroni-corrected $p < 0.05$ subset of the full DE table, further restricted to genes present in the resulting network graph. The STRING query itself duplicates the query logic of §6.1's shared helper function inline (same `required_score=700`, `species=9606` parameters).

S8.3 Integration into the 143-candidate universe

RWR was run on its own seed network (the full Bonferroni-DEP set, not restricted to the 143-candidate pool) to respect the original protocol, producing a genome/network-wide ranking. The subset of that ranking falling within the 143-candidate pool was then extracted and re-ranked locally within that 143-candidate universe (since this project's tier framework and null model, §9, are both defined on that specific universe). Of the 143 candidates, 11 have no STRING edge at ≥ 700 confidence within the RWR seed network and therefore receive no diffusion signal; this was confirmed (via the STRING ID/edge lookup) as a genuine network-topology fact rather than a script bug, and these 11 candidates are assigned rank 143 (the lowest rank in the pool) by convention, consistent with the zero-edge handling used elsewhere (§6.1).

S9 CTD tier assignment, G^* portfolio-balance metric, and null model

S9.1 CTD-based evidence tier assignment

Evidence tiers (script `RQ5_tier_audit_CTD.R`) were assigned using the Comparative Toxicogenomics Database's curated gene–disease-association file (`CTD_curated_genes_diseases.csv`, manually downloaded, filtered to rows with non-empty `DirectEvidence`). Tier 1 is defined as curated evidence for breast cancer *broadly* rather than TNBC-specifically: an audit of the actual file found curators almost never tag the narrow "Triple Negative Breast Neoplasms" MeSH term (e.g. FN1 has 0 records under that narrow term but 3 under the broad "Breast Neoplasms" term), so Tier 1 uses disease names matching `"breast"` AND a cancer keyword (`"neoplasm"` or `"carcinoma"`). Tier 2 is curated evidence for cancer broadly (a wider keyword list: neoplasm, carcinoma, sarcoma, melanoma, leukemia, lymphoma, glioma, myeloma, blastoma, mesothelioma), excluding genes already captured by Tier 1. The keyword list was audited term-by-term against the file's actual disease names to exclude false positives – `"malignant"` was dropped (matches "Malignant Hyperthermia" and "Hypertension, Malignant", neither cancer) and `"tumor"/"tumour"` was dropped (matches "Tumoral Calcinosis", a calcium-deposition disorder, and "Tumor Lysis Syndrome", a treatment complication rather than a cancer diagnosis). Tier 3 is a STRING-neighbor permutation test (below); Tier 4 is everything else.

S9.2 Tier-3 permutation test

For each of the 143 candidates, the number of its direct STRING neighbors (within a candidate-universe network built the same way as §6.1/§8.2, ≥ 700 confidence) that are themselves Tier-1 genes was counted (`n_tier1_neighbors_real`). A null distribution was built by 1,000 random permutations (`set.seed(21082026)`, distinct from the `30032026` seed used elsewhere in this pipeline) of which genes in the network are labeled "Tier 1" (same-size random draw, without replacement, from all resolved network nodes), recomputing the same neighbor count each time; a candidate's Tier-3 p-value is the proportion of permutations whose random Tier-1-neighbor count meets or exceeds its real count. A candidate is assigned Tier 3 if it is not already Tier 1/2, has at least one real Tier-1 neighbor, and this permutation $p < 0.05$.

S9.3 G^* computation

For a top-k set under a given ranking, $A/C/N$ are the within-k proportions of Tier-1, Tier-2∪Tier-3, and Tier-4 candidates respectively, and

$$G^* = \begin{cases} 0 & \text{if } A \le 0 \text{ or } C \le 0 \text{ or } N \le 0 \\ 3(A \cdot C \cdot N)^{1/3} & \text{otherwise} \end{cases}$$

. Any tier missing entirely from the top-k set forces $G^* = 0$ regardless of how the other two tiers are distributed, which is the exact mechanism behind the Equal-weight-at-k=5 case (highest Tier-1 count of any method, yet $G^* = 0$ because Tier-4 is empty) discussed in the main text. G^* was swept across $k \in \{5, 10, 15, 20, 25, 30\}$ for four rankings: AHP-CDS, FC-only, Equal-weight, and RWR (using `rank_RWR_143` from §8.3).

S9.4 Null Monte Carlo model

For each k, a null reference distribution of G^* was built via $B = 10{,}000$ draws (`set.seed(21082026)`, the same call reused for both this and the Tier-3 permutation test above – two separate procedures, not to be conflated) of k candidates sampled without replacement from the full 143-candidate tier vector (i.e. a ranking-agnostic random top-k), computing $A/C/N/G^*$ identically to §9.3 for each draw. The null's mean and 2.5th/97.5th percentiles are reported per k; this null is used descriptively as an exploratory reference band (Fig. 2D) rather than as a formal hypothesis test, and no correction for testing across the 6 depths $\times$ 4 methods is applied, consistent with that exploratory framing. $k = 25, 30$ (beyond the main text's $k \le 20$) are reported in Supplementary Table S1.

S10 Data and code availability

S10.1 Software environment

Analyses were run under R 4.5.2 (Bioconductor 3.22), with dependency versions pinned via `renv` (`renv.lock` distributed with the code repository). Key package versions: `limma` 3.66.0, `clusterProfiler` 4.18.4, `ReactomePA` 1.54.0, `biomaRt` 2.66.2, `org.Hs.eg.db` 3.22.0, `AnnotationDbi` 1.72.0, `igraph` 2.3.3, `Matrix` 1.7-4, `httr` 1.4.8, `jsonlite` 2.0.0, `dplyr` 1.2.1, `ggplot2` 4.0.3, `patchwork` 1.3.2, `ggrepel` 0.9.8, `reshape2` 1.4.5, `here` 1.0.2. The complete, exact dependency graph (all 205 pinned packages, including transitive dependencies) is given by the distributed `renv.lock` itself rather than reproduced in full here.

S10.2 External database queries

STRING (§6, §8, §9), Reactome (§4), Ensembl BioMart (§1.2, §5.2), and CTD (§9.1) were all queried in August 2026; as with any live external database, results for these resources are not guaranteed to be identical if re-queried at a later date. The one CTD file used (§9.1) was downloaded once as a static snapshot rather than queried live, so it does not carry this caveat in the same way. Github url: GitHub repository

SF Supplementary figures

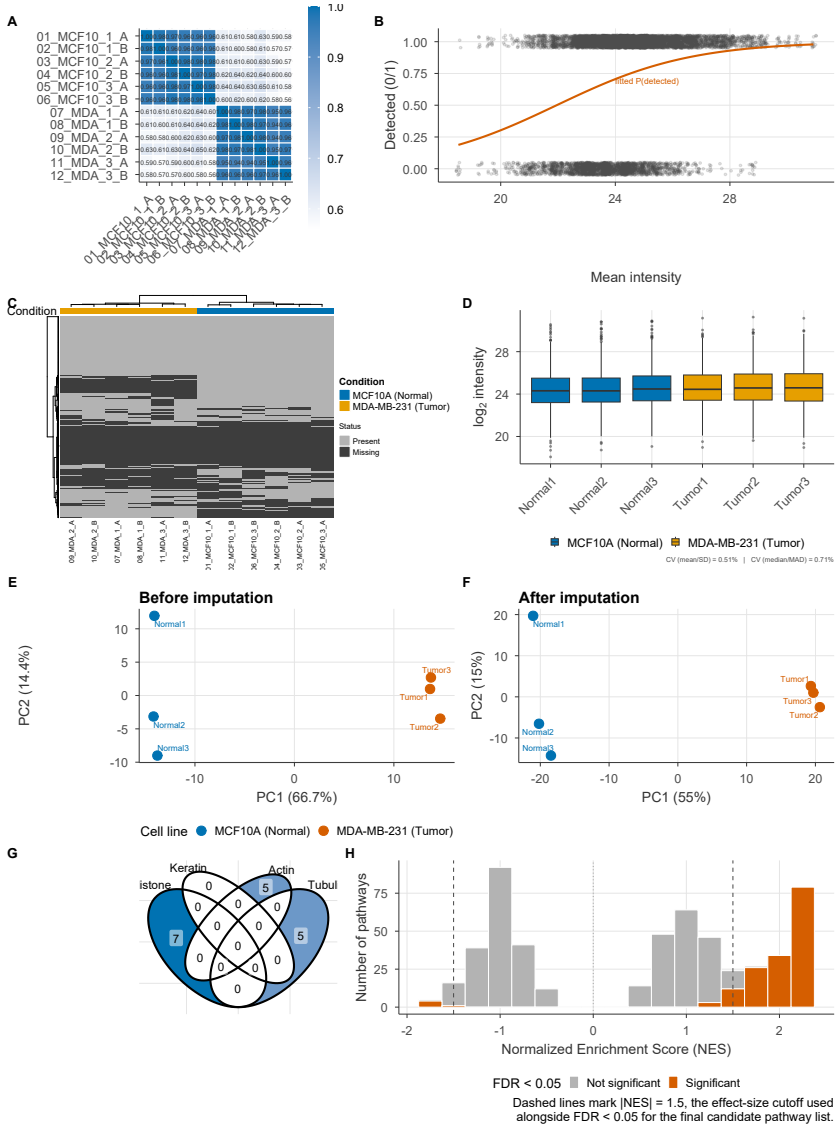


Figure S1: Quality-control and pathway-enrichment diagnostics for the exosomal proteomics dataset. **(A)** Pairwise Pearson correlation heatmap across all 12 technical replicates (samples 01–06: MCF10A; 07–12: MDA-MB-231), showing tight within-cell-line reproducibility and clear separation between cell lines. **(B)** Detection probability as a function of mean protein log-intensity, modeled by logistic regression ($\hat{\beta}_1 = 0.43$, $p < 2 \times 10^{-16}$), consistent with a limit-of-detection-driven MNAR missingness mechanism. **(C)** Missing-value pattern across all samples, annotated by cell-line condition; grey/black indicates present/missing values respectively. **(D)** Distribution of $\log_2$ intensities per sample, colored by condition (mean/SD RSD $= 0.51\%$; median/MAD RSD $= 0.71\%$), indicating comparable technical variance between conditions. **(E)** PCA before (left) and after (right) imputation, colored by cell line; PC1 scores before/after imputation are highly correlated (Pearson $r = 0.999$), indicating the shift in absolute PC1 scale reflects the change in feature set entering the covariance structure rather than an imputation artifact. **(F)** Overlap among the four content-based contaminant criteria (histone, keratin, actin, tubulin; §5.3) applied to the 193-gene core-enrichment candidate pool. No overlap between categories was observed (7 histone, 0 keratin, 5 actin, 5 tubulin; 17 genes total), consistent with these criteria targeting biologically distinct contaminant classes. The remaining 24 of the 41 total genes removed at this stage are attributable to UniProt-mapping loss and incomplete GO annotation (`na.omit()`), together yielding the 152-gene annotated pool. **(G)** Distribution of normalized enrichment scores (NES) across all Reactome pathways tested by GSEA, colored by FDR significance ($q < 0.05$; 159 pathways). Dashed lines mark $|\text{NES}| = 1.5$, the additional effect-size threshold applied to define the 153-pathway working set carried forward to core-enrichment extraction (§5.1).

ST Supplementary tables

Table S1: Portfolio-level G^* sweep, extended to $k = 25, 30$ (no method exceeds the null band at these depths).

Method	k	Tier1	Tier2/3	Tier4	G^* (null low–high)
AHP-CDS	25	6	12	7	0.955 (0.000–0.977)
AHP-CDS	30	6	14	10	0.944 (0.575–0.969)
RWR	25	2	12	11	0.770 (0.000–0.977)
RWR	30	4	15	11	0.871 (0.575–0.969)
Equal-weight	25	6	12	7	0.955 (0.000–0.977)
Equal-weight	30	6	15	9	0.932 (0.575–0.969)

Table S2: Rank-level agreement of AHP-CDS with baselines, leave-one-criterion-out (LOCO) variants, and leave-one-axis-out variants (FC+FDR: expression axis; Deg+Bet: network axis). In every variant the remaining weights are renormalized to sum to 1.

	Baselines				LOCO (criterion removed)				LOCO (axis removed)	
	FC-only	Equal-weight	Centrality-only	RWR	−FC	−FDR	−Deg	−Bet	−(FC+FDR)	−(Deg+Bet)
Spearman ρ	0.876	0.904	0.602	0.678	0.796	0.981	0.947	0.992	0.545	0.907
Kendall τ	0.684	0.736	0.454	0.487	0.607	0.886	0.798	0.927	0.388	0.729
RBO@10	0.613	0.567	0.399	0.466	0.421	0.831	0.742	0.900	0.387	0.786
RBO@15	0.642	0.603	0.399	0.475	0.435	0.848	0.745	0.907	0.384	0.771
Median Δr [Q1, Q3]	−2 [−14, 12.5]	0 [−13, 10.5]	−1 [−23.5, 19]	1 [−24, 18.5]	−2 [−18.5, 16]	0 [−5, 4.5]	0 [−8.5, 8]	0 [−4.5, 3.5]	−2 [−28, 26.5]	1 [−12, 12]
Min / max Δr	−51 / 52	−37 / 42	−78 / 86	−83 / 92	−51 / 87	−27 / 20	−29 / 35	−12 / 12	−83 / 86	−39 / 47
$\lvert\Delta r\rvert > 5$ (%)	76.2	71.3	87.4	87.4	80.4	41.3	64.3	36.4	88.8	70.6
$\lvert\Delta r\rvert > 10$ (%)	60.1	53.1	75.5	73.4	67.8	18.9	44.8	3.5	75.5	55.9
$\lvert\Delta r\rvert > 15$ (%)	44.1	40.6	63.6	62.2	56.6	6.3	26.6	0.0	66.4	39.2
$\lvert\Delta r\rvert > 20$ (%)	32.2	27.3	49.7	53.1	41.3	2.1	13.3	0.0	59.4	27.3
Top-10 shared	8/10	5/10	2/10	4/10	3/10	8/10	9/10	10/10	2/10	9/10
Top-15 shared	11/15	11/15	7/15	6/15	6/15	13/15	10/15	15/15	6/15	10/15

Note. Removing FDR alone (or Betweenness alone) left the ranking almost unchanged, whereas removing FC alone, or FC and FDR together, did not ($\rho = 0.981$ and 0.992 versus 0.796 and 0.545); in the network axis, removing Betweenness on top of Degree changed little ($\rho = 0.947$ versus 0.907; 10/15 top-15 candidates retained in both). Removing a criterion is thus largely cushioned by its axis partner, but the order of disruption also follows the order of the weights, so the table cannot separate redundancy from a small weight.

		$T = 10$					$T = 15$				
Baseline	Group	n	P_{FC}	P_{FDR}	P_{Deg}	P_{Bet}	n	P_{FC}	P_{FDR}	P_{Deg}	P_{Bet}
vs. FC-only	Promoted	43	0.260	0.194	0.403	0.142	31	0.249	0.198	0.416	0.137
	Demoted	47	0.600	0.245	0.135	0.020	33	0.647	0.246	0.097	0.010
	Stable	53	0.412	0.209	0.287	0.092	79	0.407	0.211	0.288	0.094
vs. Centrality-only	Promoted	50	0.590	0.248	0.120	0.042	43	0.607	0.245	0.110	0.038
	Demoted	61	0.263	0.171	0.433	0.133	54	0.255	0.163	0.449	0.133
	Stable	32	0.489	0.254	0.201	0.055	46	0.464	0.253	0.215	0.068

References

S. Durinck, P. T. Spellman, E. Birney, and W. Huber. Mapping identifiers for the integration of genomic datasets with the R/Bioconductor package biomaRt. *Nature Protocols*, 4(8):1184–1191, 2009.

J. Morales-Sanfrutos, J. Etxeberria-Ugartemendia, O. Barroso-Gomila, E. González, M. Sendino, P. Ximénez-Embún, F. García, E. Zarzuela, J. M. Falcón-Pérez, H. Peinado, and J. Muñoz. Defining the reference proteomes for small extracellular vesicles and non-vesicular components. *Nature Cell Biology*, 28:622–639, 2026. doi: 10.1038/s41556-026-01878-z.

J. Ren, L. Shang, Q. Wang, and J. Li. Ranking cancer proteins by integrating ppi network and protein expression profiles. *BioMed Research International*, 2019:3907195, 2019. doi: 10.1155/2019/3907195. URL https://doi.org/10.1155/2019/3907195.

T. Saaty, L. Vargas, and C. St. *The Analytic Hierarchy Process*. 07 2022. ISBN 978-1-4614-3597-6.

A. Subramanian, P. Tamayo, V. K. Mootha, S. Mukherjee, B. L. Ebert, M. A. Gillette, A. Paulovich, S. L. Pomeroy, T. R. Golub, E. S. Lander, and J. P. Mesirov. Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. *Proceedings of the National Academy of Sciences*, 102(43):15545–15550, 2005. doi: 10.1073/pnas.0506580102. URL https://doi.org/10.1073/pnas.0506580102.

S. Tyanova, T. Temu, P. Sinitcyn, A. Carlson, M. Y. Hein, T. Geiger, M. Mann, and J. Cox. The Perseus computational platform for comprehensive analysis of (prote)omics data. *Nature Methods*, 13(9): 731–740, 2016. doi: 10.1038/nmeth.3901. URL https://doi.org/10.1038/nmeth.3901.

J. A. Welsh, D. C. I. Goberdhan, L. O'Driscoll, E. I. Buzas, C. Blenkiron, B. Bussolati, H. Cai, D. Di Vizio, T. A. P. Driedonks, U. Erdbrügger, J. M. Falcon-Perez, Q.-L. Fu, A. F. Hill, M. Lenassi, S. K. Lim, M. G. Mahoney, S. Mohanty, A. Möller, R. Nieuwland, and J. Inal. Minimal information for studies of extracellular vesicles (MISEV2023): From basic to advanced approaches. *Journal of Extracellular Vesicles*, 13(2):e12404, 2024. doi: 10.1002/jev2.12404. URL https://doi.org/10.1002/jev2.12404.

G. Yu and Q. Y. He. Reactomepa: an r/bioconductor package for reactome pathway analysis and visualization. *Molecular BioSystems*, 12(2):477–479, 2016. doi: 10.1039/C5MB00663E. URL https://doi.org/10.1039/C5MB00663E.

E. A. Zaripov, A. Khraibah, P. Kasyanchyk, A. Radchanka, N. Hüttmann, and M. V. Berezovski. Cems metabolomic and lcms proteomic analyses of breast cancer exosomes reveal alterations in purine and carnitine metabolism. *Journal of Proteome Research*, 24(5):2505–2516, 2025. doi: 10.1021/acs.jproteome.4c00795. URL https://pubs.acs.org/doi/10.1021/acs.jproteome.4c00795.